

Qatar-1b

R-2568

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140925 · created 2026-08-26T23:43:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456925.65442 +/- 0.00038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.02
Transit depth (Rp/Rs)^2	1.98 +/- 0.56 [%]
Orbital Inclination (inc)	82.87 +/- 0.7
Airmass coefficient 1 (a1)	0.937 +/- 0.013
Airmass coefficient 2 (a2)	0.052 +/- 0.068
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	#3 - [261, 433]
Optimal Aperture	3.07
Optimal Annulus	9.72
Transit Duration (day)	0.0603 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.141 $\pm$ 0.02 vs 0.1463 (deviation 0.21 σ)
Duration fit vs published	0.0603 d vs 0.0692 d (deviation 1.22 σ)
Mid-time O-C	-0.9 min
Depth SNR	5.5
Residual scatter	1.33 %

Light curve

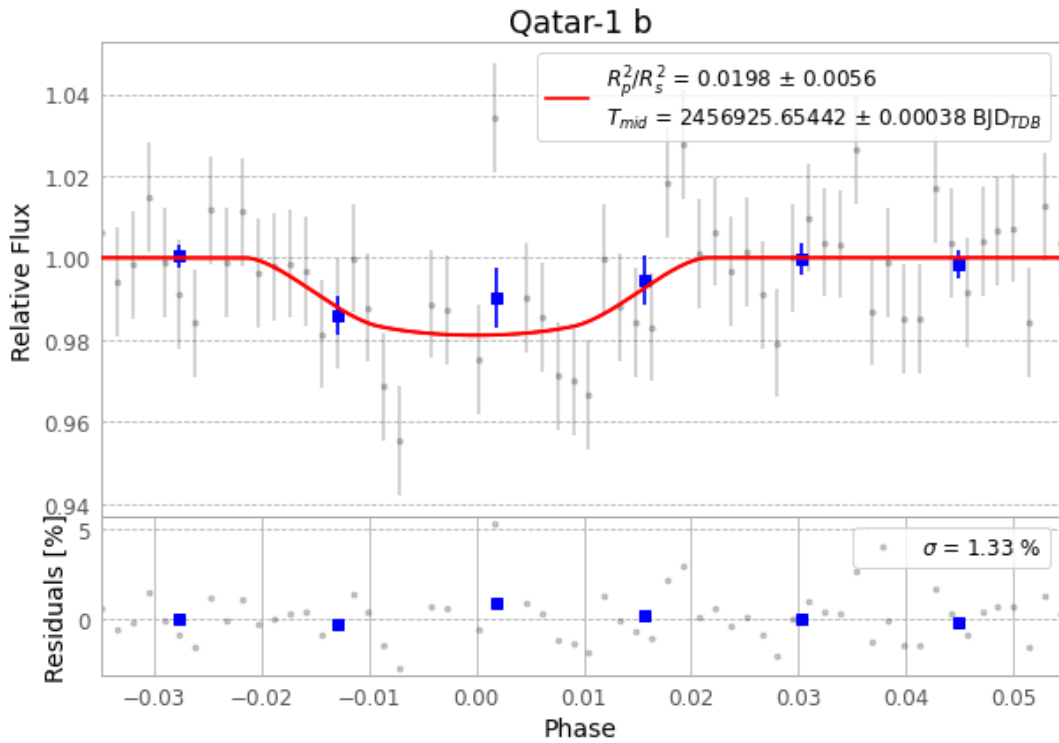


Figure 1 — FinalLightCurve_Qatar-1 b_2014-09-24.png (SHA-256 7c13d46412de757f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [307, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 10886657874f458c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

62 FITS frames (41 MB), Qatar-1140925022712.FITS → Qatar-1140925053013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140925.log (0c36c79f85a2...), FinalLightCurve_Qatar-1 b_2014-09-24.png (7c13d46412de...), FinalLightCurve_Qatar-1 b_2014-09-24.csv (72b27ba21989...)

Compute resources

Wall time 169.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:43Z.